HIGHER-ORDER CENTER CONDITIONS AND THE LIJUN–YUN CONJECTURE FOR POLYNOMIAL ABEL EQUATIONS

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ABSTRACT. We study the center problem for the Abel equation $x' = f(t)x^3 + g(t)x^2$ on $[-1,1]$, with emphasis on $g(t) = t^{n-1}$, n even. Lijun and Yun conjectured that, for $g(t) = 2t$ and f polynomial, $x = 0$ is a center if and only if f is odd; Alves, Lemos and the author proved this for $\deg f \le 5$ by showing that a center forces the first three moments $\int_{-1}^{1} fG^k\,dt$ $(k = 0, 1, 2)$ to vanish. We compute the next three center conditions in closed form: a center forces $m_3 = \frac{1}{2}\int gF^2$, $m_4 = 2\int gGF^2$ and an explicit identity for m_5, where F, G are the primitives of f, g. This explains why m_3 need not vanish at a center. For odd g vanishing only at the origin we prove that the vanishing of *all* moments is equivalent to f being odd, so the conjecture is equivalent to the statement that a center forces all moments to vanish. We give an elementary sign-change criterion that unifies and extends the known cases, a closed Cauchy-type formula for the relevant moment determinants, and a "hyperplane" theorem for $\deg f_{\mathrm{e}} \le 6$ showing that the conjecture can only fail if the odd part of f lies in an explicit affine subspace of codimension at most two. Finally, using exact Gröbner basis computations over $\mathbb{Q}$, we prove the conjecture for $g = t^{n-1}$ when $\deg f \le 7$ and $2 \le n \le 20$ (and, more generally, when $\deg f_{\mathrm{e}} \le 6$ and $\deg f_{\mathrm{o}} \le 9$), and for $g = t$ when $\deg f \le 8$.

1. INTRODUCTION

Consider the Abel differential equation

$$x' = f(t)\,x^3 + g(t)\,x^2, \qquad t \in [-1,1], \tag{1.1}$$

where f and g are real analytic on a neighbourhood of $[-1,1]$. We say that (1.1) has a *center* at $x = 0$ (on $[-1,1]$) if every solution with $|x(-1)|$ small enough satisfies $x(1) = x(-1)$. This center problem goes back to the classical Poincaré center–focus problem: in polar coordinates, and after Cherkas' transformation [12], a planar system $\dot{x} = -y + P_n(x,y)$, $\dot{y} = x + Q_n(x,y)$ with homogeneous nonlinearities becomes a trigonometric Abel equation, and its centers correspond to centers of the latter. The polynomial version on an interval was proposed by Briskin, Françoise and Yomdin as a model of the classical problem [6, 7, 8]; see also [23] for the related problem of counting solutions with $x(0) = x(1)$.

Throughout we write

$$F(t) = \int_{-1}^{t} f, \qquad G(t) = \int_{-1}^{t} g, \qquad m_k = \int_{-1}^{1} f(t)G(t)^k\,dt \quad (k \ge 0),$$

and $f = f_{\mathrm{e}} + f_{\mathrm{o}}$ for the decomposition into even and odd parts.

Composition and moments. The basic sufficient condition is the *composition condition* of Alwash and Lloyd [4]: if $F = \tilde{F} \circ W$ and $G = \tilde{G} \circ W$ for some W with $W(-1) = W(1)$, then (1.1) has a center. For small degrees and many special classes it is also necessary [6, 7, 8, 13, 3, 18], but not in general: counterexamples are known for trigonometric [3, 19] and,

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more recently, for polynomial coefficients [20]. If f is replaced by εf, the first-order condition in ε is the vanishing of all moments m_k [7]; the moments do not force composition [24], the polynomial moment problem was solved by Pakovich and Muzychuk [25], and Pakovich proved that $x' = gx^2 + \varepsilon f x^3$ has a center for all ε if and only if the composition condition holds [26]. Higher-order ("double") moments, iterated integrals and the algebraic geometry of the center set are studied in [11, 10, 14, 15, 9, 17, 1].

The Lijun–Yun conjecture. Lijun and Yun [22, Thm. 5.5] proved that $x' = 2tx^2 + \varepsilon f(t)x^3$ has a center for all small ε if and only if the polynomial f is odd (this also follows from [26]), and asked whether this remains true for a fixed ε:

Conjecture 1.1 (Lijun–Yun [22, Rem. 5.6])**.** *Let f be a polynomial. The equation $x' = 2t\,x^2 + f(t)\,x^3$, $t \in [-1,1]$, has a center at $x = 0$ if and only if f is odd.*

Since $x \mapsto \lambda x$ replaces (f, g) by $(\lambda^2 f, \lambda g)$, the coefficient $2t$ may be replaced by ct, $c \neq 0$. Following [2], we consider more generally $g(t) = t^{n-1}$ with n even. The "if" part is the composition condition with $W(t) = t^2$. For the "only if" part, Alves, Lemos and the author [2] proved that a center of (1.1) forces $m_0 = m_1 = m_2 = 0$, and deduced the conjecture when $\deg f \leq 5$, or more generally when g is odd, vanishes only at 0, and the even part of f changes sign at most twice on $[0,1]$. They also observed, using an example of Giné, Grau and Santallusia [19], that a center need not satisfy $m_3 = 0$.

Main results. Our first result computes the next center conditions in closed form. Put $H(t) = \int_{-1}^{t} fG$.

Theorem A (Higher-order center conditions)**.** *Assume $G(1) = 0$. If (1.1) has a center at $x = 0$, then $m_0 = m_1 = m_2 = 0$ and*

$$m_3 = \frac{1}{2}\int_{-1}^{1} g\,F^2\,dt, \tag{1.2}$$

$$m_4 = 2\int_{-1}^{1} g\,G\,F^2\,dt, \tag{1.3}$$

$$m_5 = \frac{1}{2}\int_{-1}^{1} g\,F^3\,dt + \frac{9}{2}\int_{-1}^{1} g\,G^2F^2\,dt - \int_{-1}^{1} f\,G^2H\,dt. \tag{1.4}$$

For the example of [19] (on $[0, 2\pi]$), both sides of (1.2) equal $\pi/2$, which explains the observation of [2]; see Example 3.2.

From now on we say that g satisfies *Condition I* if g is odd and vanishes only at $t = 0$ [2]; $g(t) = t^{n-1}$, n even, is the model case.

Theorem B (Reformulation of the conjecture)**.** *Let g satisfy Condition I and let f be continuous. Then $m_k = 0$ for all $k \geq 0$ if and only if f is odd. Consequently, for such g and polynomial (or analytic) f, the following are equivalent:*

(i) *every center of (1.1) has f odd;*
(ii) *every center of (1.1) has $m_k = 0$ for all k.*

Theorem B shows that, for $g = t^{n-1}$, Conjecture 1.1 is precisely the statement "center $\Rightarrow$ moment conditions" (an implication which is open in general; cf. the diagram in [2, §1.3]). The proof is elementary and rests on a sign-change lemma (Lemma 5.1). The same lemma, combined with Theorem A, gives the following criterion. For $\varphi, \psi \in L^2(0,1)$ and a weight w put $\langle \varphi, \psi \rangle_w = \int_0^1 w\varphi\psi\,dt$, and set

$$A(t) = \int_0^t f_{\mathrm{e}}, \qquad C(t) = \int_t^1 f_{\mathrm{o}} \qquad (0 \leq t \leq 1).$$

Theorem C (Sign-change criterion)**.** *Let g satisfy Condition I and let f be a polynomial. Write $f_{\mathrm{e}}(t) = p(t^2)$. In each of the following cases, (1.1) has a center at $x = 0$ if and only if f is odd:*

(a) *p changes sign at most twice in $(0,1)$;*
(b) *p changes sign at most three times in $(0,1)$ and $\langle A, C\rangle_g = 0$;*
(c) *p changes sign at most four times in $(0,1)$ and $\langle A, C\rangle_g = \langle A, C\rangle_{gG} = 0$.*

Case (a) is [2, Thm. 1.10], here with a short self-contained proof; it contains [2, Thm. 1.12] and the case $\deg f_{\mathrm{e}} \le 4$ with f_{o} arbitrary. Since A depends on f_{e}, the orthogonality conditions in (b) and (c) are implicit. For $\deg f_{\mathrm{e}} \le 6$ they can be replaced by explicit conditions on f_{o} only:

Theorem D (Hyperplane theorem)**.** *Let g satisfy Condition I with $g > 0$ on $(0,1]$. There is a nonzero even polynomial e of degree at most 6, unique up to a constant factor, with $\int_{-1}^{1} eG^k = 0$ for $k = 0, 1, 2$; moreover $\mu := \int_{-1}^{1} eG^3 \neq 0$. Let $\nu := \int_{-1}^{1} eG^4$ and $A_e(t) = \int_0^t e$. If f is a polynomial with $\deg f_{\mathrm{e}} \le 6$ and (1.1) has a center, then either f is odd, or $f_{\mathrm{e}} = a\,e$ with $a \neq 0$ and*

$$\langle A_e, C\rangle_g = -\frac{\mu}{2}, \qquad \langle A_e, C\rangle_{gG} = -\frac{\nu}{8}. \tag{1.5}$$

In particular, if the odd part of f violates either equality in (1.5), then (1.1) has a center if and only if f is odd.

Since C depends linearly on f_{o}, (1.5) defines an affine subspace of codimension at most two in the space of odd parts. Finally, the higher center conditions can be handled by exact computer algebra:

Theorem E (Computer-assisted)**.** *Let $g(t) = t^{n-1}$.*

(i) *For every even n with $2 \le n \le 20$ and every polynomial f with $\deg f_{\mathrm{e}} \le 6$ and $\deg f_{\mathrm{o}} \le 9$, (1.1) has a center if and only if f is odd; for $n \in \{2, 4, 6\}$ the same holds with $\deg f_{\mathrm{o}} \le 11$. In particular Conjecture 1.1 (and its version with t^{n-1}, $2 \le n \le 20$) holds whenever $\deg f \le 7$.*
(ii) *For $n = 2$, Conjecture 1.1 holds whenever $\deg f \le 8$.*

The proof of Theorem E computes the first K center conditions as exact polynomials in the coefficients of f and shows, by Gröbner bases over $\mathbb{Q}$, that they have no common complex zero with $f_{\mathrm{e}} \neq 0$ (Section 7). Interestingly, the number of conditions needed is sharp: for $\deg f \le 7$ the first 8 conditions admit complex common zeros with $f_{\mathrm{e}} \neq 0$ for every n tested, whereas 9 conditions exclude them.

The paper is organized as follows. Section 2 sets up the expansion of the return map. Theorem A is proved in Section 3. Section 4 contains the parity reduction, Section 5 the moment lemmas and Theorem B, and Section 6 the proofs of Theorems C and D. Section 7 describes the computations behind Theorem E, and Section 8 lists further remarks and open problems.

2. The expansion of the return map

Let $x(t, \rho)$ be the solution of (1.1) with $x(-1, \rho) = \rho$.

Lemma 2.1. *For $|\rho|$ small, the problem*

$$u' = g(t) + \frac{\rho\, f(t)}{1 - \rho u}, \qquad u(-1) = 0, \tag{2.1}$$

has a solution $u(t,\rho)$ on $[-1,1]$, analytic in ρ, and $x(t,\rho)=\rho/(1-\rho u(t,\rho))$. In particular, (1.1) has a center at $x=0$ if and only if $u(1,\rho)\equiv 0$, i.e. if and only if $u_k(1)=0$ for all $k\geq 0$, where $u=\sum_{k\geq 0}u_k\rho^k$.

Proof. For $\rho=0$, (2.1) has the solution $u=G$ on $[-1,1]$; the right-hand side is analytic in (u,ρ) near $\{(G(t),0)\}$, so by analytic dependence on parameters $u(t,\rho)$ exists on $[-1,1]$ and is analytic in ρ for $|\rho|$ small. The function $y=\rho/(1-\rho u)$ satisfies $y(-1)=\rho$ and $y'=\rho^2u'/(1-\rho u)^2=y^2(g+fy)$, so $y=x$ by uniqueness. For $\rho\neq 0$, $x(1,\rho)=\rho$ is equivalent to $u(1,\rho)=0$. □

Writing $S:=(1-\rho u)^{-1}=\sum_{k\geq 0}S_k\rho^k$, the identity $S=1+\rho uS$ and (2.1) give the recursion

$$u_0=G,\qquad u_k(t)=\int_{-1}^{t} f\,S_{k-1}\quad(k\geq 1),\qquad S_0=1,\qquad S_k=\sum_{i=0}^{k-1}u_i\,S_{k-1-i}. \tag{2.2}$$

Explicitly,

$$\begin{aligned}
&S_1=u_0,\qquad S_2=u_1+u_0^2,\qquad S_3=u_2+2u_0u_1+u_0^3,\\
&S_4=u_3+2u_0u_2+u_1^2+3u_0^2u_1+u_0^4,\\
&S_5=u_4+2u_0u_3+2u_1u_2+3u_0^2u_2+3u_0u_1^2+4u_0^3u_1+u_0^5.
\end{aligned}$$

In particular $u_0(1)=G(1)$, so $G(1)=0$ is necessary for a center. When f and g are polynomials, every u_k is a polynomial in t whose coefficients are polynomials with rational coefficients in those of f and g.

Remark 2.2. The expansion in powers of ρ of [22, 2] uses $x=\rho+\sum r_k\rho^k$; the variable u is related by $1/x=1/\rho-u$. Its advantage is that $u_{k+1}(1)$ contains the moment m_k *linearly*, with corrections that are at least quadratic in f (Theorem A).

3. Proof of Theorem A

Let $K_2(t)=\int_{-1}^{t}fG^2$. By (2.2),

$$u_1=F,\qquad u_2=H,\qquad u_3=\tfrac12F^2+K_2,$$

hence $u_1(1)=m_0$, $u_2(1)=m_1$ and $u_3(1)=\frac12m_0^2+m_2$. A center thus forces $m_0=m_1=m_2=0$ (this is [2, Thm. 1.7]). From now on assume $G(1)=m_0=m_1=m_2=0$, so that

$$F(\pm1)=G(\pm1)=H(\pm1)=K_2(\pm1)=0.$$

All integrals below are over $[-1,1]$, and integrations by parts produce no boundary terms.

Order 4. By (2.2), $u_4(1)=\int f(H+2GF+G^3)$. Since $F'=f$, $\int fH=-\int FfG$, and since $(F^2)'=2fF$, $\int fFG=-\frac12\int gF^2$. Hence

$$u_4(1)=m_3+\int fFG=m_3-\frac12\int gF^2. \tag{3.1}$$

Order 5. $u_5(1)=\int fS_4$. We have $\int fF^2=\frac13F^3\big|_{-1}^{1}=0$, $\int fK_2=-\int FfG^2$, $2\int fGH=2\int HH'=0$ and $\int fG^4=m_4$. Therefore

$$u_5(1)=m_4+2\int fFG^2=m_4+\int G^2(F^2)'=m_4-2\int gGF^2. \tag{3.2}$$

Order 6. $u_6(1)=\int fS_5$. Integrating by parts, $\int fu_4=-\int Ff(H+2GF+G^3)$, so

$$u_6(1)=m_5+\int fFH+2\int fGF^2+3\int fG^3F+2\int fGK_2+3\int fG^2H.$$

Now $\int fGK_2 = \int H'K_2 = -\int HfG^2$, $\int fFH = \int F'FH = -\frac{1}{2}\int fGF^2$, $\int fGF^2 = \frac{1}{3}\int G(F^3)' = -\frac{1}{3}\int gF^3$ and $\int fG^3F = \frac{1}{2}\int G^3(F^2)' = -\frac{3}{2}\int gG^2F^2$. Collecting terms,

$$u_6(1) = m_5 - \frac{1}{2}\int gF^3 - \frac{9}{2}\int gG^2F^2 + \int fG^2H. \tag{3.3}$$

By Lemma 2.1, a center gives $u_4(1) = u_5(1) = u_6(1) = 0$, which is (1.2)–(1.4). □

Remark 3.1. The proof of (3.1) only uses $m_0 = m_1 = 0$. The identities (3.1)–(3.3) hold on any interval $[a, b]$ with $G(a) = G(b) = 0$, with the obvious changes. We also verified them symbolically for generic polynomials of degree 7 and $g = t^{n-1}$, $n = 2, 4, 6$, and checked the recursion (2.2) against high-precision numerical integration of (1.1).

Example 3.2. The trigonometric Abel equation

$$\frac{dx}{d\theta} = (\sin\theta - \sin 2\theta + \sin 3\theta)x^3 + (\cos\theta + 2\cos 2\theta)x^2, \qquad \theta \in [0, 2\pi],$$

has a center [19], and $m_0 = m_1 = m_2 = 0$, $m_3 = \pi/2$ [2, Rem. 1.8]. Here $G = \sin\theta + \sin 2\theta$ vanishes at 0 and 2π, and a direct symbolic computation gives $\frac{1}{2}\int_0^{2\pi} gF^2\,d\theta = \pi/2$, in agreement with (1.2). Thus $m_3 \neq 0$ at this center is not an anomaly: the fourth center condition is $m_3 = \frac{1}{2}\int gF^2$, not $m_3 = 0$.

Remark 3.3. The integrals in (1.2)–(1.4) are iterated integrals of $f\,dt$ and $g\,dt$; e.g. $\int fG^2H = \iint_{-1<s<t<1} f(s)G(s)\,f(t)G(t)^2\,ds\,dt$. Theorem A is therefore an explicit instance of Brudnyi's expansion of the first return map in iterated integrals [11], and of the second-order ("double moment") conditions studied in [14, 9], in a normalization in which the moments appear linearly.

4. Parity

Assume that g satisfies Condition I. Replacing x by $-x$ changes (f, g) into $(f, -g)$, so we may assume $g > 0$ on $(0, 1]$. Then G is even, $G(\pm 1) = 0$, $G < 0$ on $(-1, 1)$, and G restricted to $[0, 1]$ is an increasing bijection onto $[G(0), 0]$; let $\tau : [G(0), 0] \to [0, 1]$ be its inverse. Since fG^k has the same parity as f, only f_{e} contributes to the moments:

$$m_k = \int_{-1}^{1} f_{\mathrm{e}}\,G^k\,dt = 2\int_0^1 f_{\mathrm{e}}\,G^k\,dt. \tag{4.1}$$

Lemma 4.1. *Assume $m_0 = 0$. For $t \in [0, 1]$,*

$$F(t) = A(t) - C(t), \qquad F(-t) = -A(t) - C(t).$$

Consequently, for every $j \geq 0$,

$$\int_{-1}^{1} gG^jF^2\,dt = -4\langle A, C\rangle_{gG^j}, \qquad \int_{-1}^{1} gG^jF^3\,dt = 2\int_0^1 gG^j\,(A^3 + 3AC^2)\,dt.$$

Proof. By (4.1), $\int_0^1 f_{\mathrm{e}} = 0$. For $t \in [0, 1]$, $\int_{-1}^t f_{\mathrm{e}} = \int_0^t f_{\mathrm{e}} = A(t)$ and $\int_{-1}^t f_{\mathrm{o}} = -\int_t^1 f_{\mathrm{o}} = -C(t)$, which gives $F(t)$; the formula for $F(-t)$ follows from the parities of $f_{\mathrm{e}}, f_{\mathrm{o}}$. Since gG^j is odd, $\int_{-1}^1 gG^j\Phi(F) = \int_0^1 gG^j[\Phi(F(t)) - \Phi(F(-t))]dt$ for any Φ, and $(A - C)^2 - (A + C)^2 = -4AC$, $(A - C)^3 + (A + C)^3 = 2A^3 + 6AC^2$. □

Similarly, if $m_1 = 0$, then $H(t) = B(t) - D(t)$ and $H(-t) = -B(t) - D(t)$ on $[0, 1]$, where $B(t) = \int_0^t f_{\mathrm{e}}G$ and $D(t) = \int_t^1 f_{\mathrm{o}}G$, and a computation as above gives $\int_{-1}^1 fG^2H = 2\int_0^1 G^2(f_{\mathrm{o}}B - f_{\mathrm{e}}D)$. Combining with Theorem A:

Corollary 4.2. *Let g satisfy Condition I. If (1.1) has a center, then $m_0 = m_1 = m_2 = 0$ and*

$$m_3 = -2\langle A, C\rangle_g, \qquad m_4 = -8\langle A, C\rangle_{gG},$$
$$m_5 = \int_0^1 g(A^3 + 3AC^2) - 18\langle A, C\rangle_{gG^2} - 2\int_0^1 G^2(f_{\mathrm{o}}B - f_{\mathrm{e}}D).$$

Remark 4.3. In the variable $s = G(t)$ one has $g\,dt = ds$, so $\langle A, C\rangle_{gG^j} = \int_{G(0)}^0 (AC)(\tau(s))\, s^j\, ds$: the fourth and fifth center conditions say that the 0-th and 1-st moments of AC in the variable G are determined by m_3 and m_4, the corresponding moments of f_{e} (cf. Proposition 5.2). For $g = t^{n-1}$ the weight t^{n-1} on $(0,1)$ is a shifted Jacobi weight, and $\langle A, C\rangle_g$ can be expanded in the polynomials $P_k^{(0,n-1)}(2t-1)$; in particular, disjoint Jacobi spectral supports of A and C imply $\langle A, C\rangle_g = 0$.

5. Moments and sign changes

We say that $\varphi \in L^1(\alpha, \beta)$ has at most N sign changes if there are $\alpha = s_0 < s_1 < \cdots < s_r < s_{r+1} = \beta$, $r \le N$, such that $(-1)^i\varphi \ge 0$ a.e. on (s_i, s_{i+1}), or $(-1)^i\varphi \le 0$ a.e. on all of them.

Lemma 5.1. *If $\varphi \in L^1(\alpha, \beta)$ has at most N sign changes and $\int_\alpha^\beta \varphi(s)s^k\, ds = 0$ for $k = 0, \dots, N$, then $\varphi = 0$ a.e.*

Proof. With $s_1, \dots, s_r$ as above, $\varphi(s)\prod_{i=1}^r (s - s_i)$ has constant sign and, being φ times a polynomial of degree $r \le N$, has zero integral. Hence it vanishes a.e., and so does φ. (This is the classical Chebyshev-system argument; see [21, 27].) □

Proposition 5.2. *Let g satisfy Condition I with $g > 0$ on $(0,1]$, and let $f_{\mathrm{e}}(t) = p(t^2)$ with p continuous on $[0,1]$. Then*

$$m_k = 2\int_{G(0)}^0 \varphi(s)\, s^k\, ds, \qquad \varphi(s) = \frac{p(\tau(s)^2)}{g(\tau(s))},$$

where $\varphi \in L^1(G(0), 0)$ has the same sign changes as p on $(0,1)$. In particular, if p has at most N sign changes in $(0,1)$ and $m_0 = \cdots = m_N = 0$, then $p \equiv 0$.

Proof. Substitute $s = G(t)$ in (4.1). Integrability holds because $\int |\varphi|\, ds = \int_0^1 |p(t^2)|\, dt$; the singularity of $1/g$ at $t = 0$ is compensated by $ds = g\,dt$. Since $g > 0$ and τ is increasing, φ and $p \circ \tau^2$ have the same sign changes. Apply Lemma 5.1. □

Proof of Theorem B. If f is odd, fG^k is odd and $m_k = 0$. Conversely, if all m_k vanish, then by Proposition 5.2 all moments of $\varphi \in L^1$ on the compact interval $[G(0), 0]$ vanish; by the Weierstrass approximation theorem, $\varphi = 0$ a.e., i.e. $f_{\mathrm{e}} \equiv 0$. The equivalence of (i) and (ii) follows. □

Remark 5.3. For $g = t^{n-1}$ the first part of Theorem B can also be derived from [25]: up to affine changes of variable, the right compositional factors of $G = (t^n - 1)/n$ are t^d with $d \mid n$, and $W(-1) = W(1)$ forces d to be even. Note also that $\{t^{nk}\}_{k\ge 0}$ is complete in $C[0,1]$ by the Müntz theorem [5], since $\sum 1/(nk) = \infty$.

For $g = t^{n-1}$ the moment matrices have a closed-form determinant, which generalizes the 3×3 determinant computed in [2, Thm. 1.12].

Proposition 5.4. *Let $n \ge 2$ be even, $G(t) = (t^n - 1)/n$ and $M^{(N)} = \left(\int_{-1}^1 t^{2j}G(t)^k\, dt\right)_{k,j=0}^N$. Then*

$$\det M^{(N)} = \frac{2^{N+1}}{n^{N(N+1)/2}} \cdot \frac{\prod_{0\le i<k\le N} 2n\,(k-i)^2}{\prod_{i,j=0}^N (ni + 2j + 1)} > 0.$$

Proof. Expanding $(t^n-1)^k$, row k of $M^{(N)}$ equals n^{-k} times row k of $\big(2/(ni+2j+1)\big)_{i,j}$ plus a combination of the previous rows. Hence $\det M^{(N)} = 2^{N+1}n^{-N(N+1)/2}\det\big(1/(x_i+y_j)\big)$ with $x_i = ni+1$, $y_j = 2j$, and the Cauchy determinant equals $\prod_{i<k}(x_k-x_i)(y_k-y_i)/\prod_{i,j}(x_i+y_j)$. □

Remark 5.5. Proposition 5.4 shows that $m_0 = \cdots = m_N = 0$ means $f_{\mathrm{e}} \perp t^{nk}$ in $L^2(-1,1)$ for $k = 0,\dots,N$, an orthogonality problem of Müntz type [5]. For $n = 2$ this is orthogonality to all even polynomials of degree $\le 2N$; e.g. for $\deg f_{\mathrm{e}} \le 6$ and $N = 2$, f_{e} is a multiple of the Legendre polynomial P_6.

6. Proofs of Theorems C and D

Proof of Theorem C. If f is odd, write $f(t) = t\hat{f}(t^2)$, $g(t) = t\hat{g}(t^2)$; then F and G are functions of $W(t) = t^2$, with $W(-1) = W(1)$, and the composition condition [4] gives a center. Conversely, assume a center; we may assume $g > 0$ on $(0,1]$. By Corollary 4.2, $m_0 = m_1 = m_2 = 0$ and, under the orthogonality hypotheses of (b) (resp. (c)), also $m_3 = 0$ (resp. $m_3 = m_4 = 0$). Proposition 5.2 with $N = 2,3,4$ gives $p \equiv 0$. □

In particular, Theorem C(b) contains the statement: if $g = t^{n-1}$, $\deg f_{\mathrm{e}} \le 6$ and A, C are orthogonal in $L^2((0,1), t^{n-1}dt)$, then a center forces f to be odd.

Proof of Theorem D. Let V be the 4-dimensional space of even polynomials of degree ≤ 6. By Proposition 5.2 with $N = 3$ (a polynomial of degree ≤ 3 in $y = t^2$ has at most three sign changes), the linear map $V \to \mathbb{R}^4$, $f_{\mathrm{e}} \mapsto (m_0,\dots,m_3)$, is injective, hence bijective. Therefore the kernel of (m_0, m_1, m_2) on V is spanned by a single $e \neq 0$, and $\mu = m_3(e) \neq 0$. If (1.1) has a center and $\deg f_{\mathrm{e}} \le 6$, then $m_0 = m_1 = m_2 = 0$, so $f_{\mathrm{e}} = ae$ and $A = aA_e$. By Corollary 4.2,

$$a\mu = m_3 = -2a\langle A_e, C\rangle_g, \qquad a\nu = m_4 = -8a\langle A_e, C\rangle_{gG}.$$

If $a \neq 0$, this is (1.5). □

Remark 6.1. The orthogonality hypothesis $\langle A, C\rangle_g = 0$ of Theorem C(b) avoids the first hyperplane in (1.5), since $\mu \neq 0$; Theorem D shows that it suffices to avoid either one. On the exceptional set, the sixth center condition (Corollary 4.2) becomes a quadratic equation in (a, f_{o}) with a term $a^2\int_0^1 gA_e^3$; this is the starting point of the computations of Section 7.

Example 6.2. Let $f_{\mathrm{o}} = b_0t + b_1t^3 + b_2t^5 + b_3t^7$. For $g = t$ $(n = 2)$, $e(t) = 231t^6 - 315t^4 + 105t^2 - 5 = 16P_6(t)$ and, with $f_{\mathrm{e}} = a\,e/231$,

$$u_4(1) = \tfrac{64a}{530675145}\,(255b_1 + 238b_2 + 189b_3 + 765),$$
$$u_5(1) = \tfrac{256a}{10082827755}\,(4845b_0 + 969b_1 + 133b_2 - 63b_3 - 3876).$$

For $g = t^3$ $(n = 4)$, $e(t) = 385t^6 - 585t^4 + 231t^2 - 15$ and, with $f_{\mathrm{e}} = a\,e/385$,

$$u_4(1) = -\tfrac{64a}{5601570975}\,(1615b_0 - 399b_2 - 462b_3 - 693),$$
$$u_5(1) = \tfrac{256a}{128836132425}\,(16169b_0 + 5313b_1 + 1817b_2 + 561b_3 - 2376).$$

In both cases the exceptional set (1.5) is a 2-dimensional affine plane in $\mathbb{R}^4$.

7. Computer-assisted proof of Theorem E

Fix n and degrees $2E$, $2R+1$, and write

$$f(t) = \sum_{j=0}^{E} a_j t^{2j} + \sum_{j=0}^{R} b_j t^{2j+1}, \qquad g(t) = t^{n-1}.$$

By (2.2), each $c_k := u_k(1)$ is a polynomial in $\mathbb{Q}[a_0, \dots, a_E, b_0, \dots, b_R]$, and by Lemma 2.1, a center forces $c_1 = \cdots = c_K = 0$ for every K. Let I_K be the ideal generated by $c_1, \dots, c_K$. If, for each i,

$$1 \in I_K + \langle 1 - z\, a_i \rangle \subset \mathbb{Q}[z, a, b], \tag{7.1}$$

then every common complex zero of $c_1, \dots, c_K$ has $a_i = 0$ (Rabinowitsch's trick), so a center forces $f_{\mathrm{e}} = 0$, and Theorem C's sufficiency part finishes the proof. Condition (7.1) holds if and only if the reduced Gröbner basis of $I_K + \langle 1 - z a_i \rangle$ is $\{1\}$.

Reduction of variables. By Theorem A, c_1, c_2, c_3 vanish if and only if $m_0 = m_1 = m_2 = 0$, a linear system in $a_0, \dots, a_E$ of rank 3 (Proposition 5.4). We therefore parametrize its solution space as $f_{\mathrm{e}} = \sum_{i=1}^{E-2} \gamma_i e_i$, with an explicit basis e_i of integer polynomials, check that c_1, c_2, c_3 vanish identically on this family, and apply (7.1) to $c_4, \dots, c_K$ and the variables γ_i. This is equivalent to the unreduced test, and much faster.

Implementations. The polynomials c_k were computed by two independent implementations of (2.2): one in `SymPy` (exact rational arithmetic) and one written directly in `Singular` [16]. We checked that both give identical values of $c_1, \dots, c_6$ at random rational parameters, and that the truncated series $\rho/(1 - \rho \sum_{k \le 8} c_k \rho^k)$ agrees with the numerical solution of (1.1) up to the expected order $O(\rho^{11})$. Gröbner bases were computed with `SymPy`'s `groebner` and with `Singular`'s `std` (degree reverse lexicographic order, characteristic 0). All rows of Table 1 were obtained with `Singular`; the rows with $\deg f_{\mathrm{e}} \le 6$ were also reproduced with `SymPy` for $n \in \{2, 4, 6\}$ (and $\deg f_{\mathrm{o}} \le 7$), and for $n = 2$, $\deg f_{\mathrm{o}} \le 9$, with identical outcomes. The code is available in the repository cited in the Data availability statement.

n	$\deg f_{\mathrm{e}} \le$	$\deg f_{\mathrm{o}} \le$	K	common zeros of $c_1, \dots, c_K$ with $f_{\mathrm{e}} \neq 0$
$2, 4, \dots, 20$	6	7	8	exist (over $\mathbb{C}$)
$2, 4, \dots, 20$	6	7	9	none
$2, 4, \dots, 20$	6	9	9	exist (over $\mathbb{C}$)
$2, 4, \dots, 20$	6	9	10	none
$2, 4, 6$	6	11	10	exist (over $\mathbb{C}$)
$2, 4, 6$	6	11	11	none
2	8	7	11	none

TABLE 1. Results of the exact Gröbner basis computations.

Proof of Theorem E. For (i), use the rows of Table 1 marked "none" with $\deg f_{\mathrm{e}} \le 6$. For (ii), use the last row: a polynomial of degree ≤ 8 has $\deg f_{\mathrm{e}} \le 8$ and $\deg f_{\mathrm{o}} \le 7$. In both cases a center forces $c_1 = \cdots = c_K = 0$, hence $f_{\mathrm{e}} = 0$, and conversely odd f gives a center. □

Remark 7.1. (1) The computations are exact and decide the absence of *complex* solutions; the conclusion therefore holds a fortiori over $\mathbb{R}$. (2) The rows with "exist" show that, in these degrees, the first $K - 1$ conditions do not suffice. In particular, the sixth to eighth center conditions alone do not exclude the exceptional set of Theorem D. (3) The computations for $\deg f_{\mathrm{e}} \le 6$ take seconds (minutes for $\deg f_{\mathrm{o}} \le 11$); the case $\deg f_{\mathrm{e}} \le 8$ took about one hour of a single core with `std`. A modular computation (`modStd`) suggests that already $K = 10$ suffices in that case, but we only claim what was obtained with `std` over $\mathbb{Q}$.

8. Further remarks and open problems

Problem 8.1. Prove Conjecture 1.1 for $\deg f_{\mathrm{e}} \le 6$ and f_{o} of arbitrary degree. By Theorem D, it suffices to show that the higher center conditions exclude the exceptional set (1.5). In all cases of Table 1 with $\deg f_{\mathrm{e}} \le 6$ and $\deg f_{\mathrm{o}} = 2R+1$, exactly $K = R+6$ conditions are needed ($K = 9, 10, 11$ for $R = 3, 4, 5$). Is this true for all R?

Problem 8.2. Find a closed formula for $u_{k+1}(1)$, modulo the previous conditions, of the form $m_k + \mathcal{Q}_k$, where $\mathcal{Q}_k$ is a combination of iterated integrals of order ≥ 2 in f. Theorem A gives $\mathcal{Q}_3, \mathcal{Q}_4, \mathcal{Q}_5$; Brudnyi's formalism [11] suggests that a general expression exists. Together with Lemma 4.1, it would extend Theorem C to arbitrary numbers of sign changes, under additional orthogonality hypotheses.

Problem 8.3. Extend Theorems C–E to trigonometric Abel equations with a reflection symmetry $\theta \mapsto -\theta$, where Example 3.2 shows that the nonlinear correction in (1.2) is essential.

Data availability

The `SymPy` and `Singular` code used for the identities of Theorem A, the determinant formula of Proposition 5.4, Example 6.2 and Table 1 is available at `https://github.com/Araujo2021/araujo2021.github.io/tree/main/pesquisa/abel-lijun-yun/codigo`.

Acknowledgements

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